

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Application Based Questions
Weightage:	40% of CIA	Total Marks:	100 (2 Questions × 50 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
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Section / Batch:	3 rd (AI&DS)	Date:	15-07-2026

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Clearly show all steps in derivations and complexity analysis.
- Use proper asymptotic notation (Big-O, Big-Θ, Big-Ω) wherever required.
- Justify each step in recurrence solving using appropriate methods.
- Neat presentation and logical explanation are mandatory

Question Type	Focus Area	Questions	Marks
Application-Based Questions	Apply Master Theorem and asymptotic analysis to real-world scenarios	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – APPLICATION BASED QUESTIONS

Q1. Search Engine Indexing A search engine indexing algorithm is defined as: $T(n) = 2T(n/2) + n^2$. Apply the Master Theorem and determine the time complexity. Clearly explain the chosen case. Analyze how indexing time grows with web pages.

Q2. AI-Based Chatbot Processing An AI chatbot processes user queries using: $T(n) = T(n/2) + \log n$. Apply the Master Theorem and determine time complexity. Identify parameters and justify the case. Discuss its impact on response time as user queries increase.

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Question No.	Understanding of Problem Context (20%)	Application of Concepts (30%)	Problem-Solving Approach (20%)	Accuracy of Solution (20%)	Clarity (10%)	Total Marks
Q1 (50 Marks)						
Q2 (50 Marks)						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Q1. Search Engine Indexing

Given Recurrence

$$T(n) = 2T(n/2) + n^2$$

Step 1: Identify the Parameters

Compare with the Master Theorem form:

$$T(n) = aT(n/b) + f(n)$$

Where:

- $a = 2$ (Number of subproblems)
- $b = 2$ (Each subproblem is of size $n/2$)
- $f(n) = n^2$

Step 2: Calculate ($n^{\log_b a}$)

$$n^{\log_2 2} = n^1 = n$$

Thus,

- ($n^{\log_b a} = n$)
- ($f(n) = n^2$)

Since

$$n^2 = \Omega(n^{1+\epsilon})$$

where ($\epsilon = 1$),

the function $f(n)$ grows polynomially faster than $n^{\log_b a}$

Step 3: Regularity Condition

Check:

$$af(n/b) \leq cf(n)$$

$$2 \left(\frac{n}{2} \right)^2 = 2 \times \frac{n^2}{4} = \frac{n^2}{2}$$

Since

$$\frac{n^2}{2} \leq cn^2$$

For $c = 1/2 < 1$,
the regularity condition is satisfied.

Step 4: Apply Master Theorem

This satisfies Case 3 of the Master Theorem.

Therefore,

$$T(n) = \Theta(n^2)$$

Recursion Tree Verification

Level 0

$$n^2$$

Level 1

There are 2 subproblems.

Cost:

$$2(n/2)^2 = n^2/2$$

Level 2

There are 4 subproblems.

Cost:

$$4(n/4)^2 = n^2/4$$

Total Cost

$$n^2 + n^2/2 + n^2/4 + \dots$$

This is a geometric series whose sum is

$$2n^2$$

Hence,

$$\Theta(n^2)$$

Analysis of Search Engine Indexing

A search engine creates an index by scanning webpages, extracting keywords, hyperlinks, metadata, and document information. During indexing:

- Every webpage must be analyzed.
- Keywords are stored in index tables.
- Links between webpages are processed.
- Duplicate pages are removed.
- Ranking information is updated.

Since the recurrence has quadratic complexity:

- Doubling the number of webpages increases indexing work by nearly **four times**.
- Large search engines require distributed computing and parallel indexing to reduce execution time.
- Efficient indexing algorithms improve search speed and reduce server load.

Q2. AI-Based Chatbot Processing

Given Recurrence

$$T(n) = T(n/2) + \log n$$

Step 1: Identify Parameters

Compare with

$$T(n) = aT(n/b) + f(n)$$

Here,

- $a = 1$
- $b = 2$

- $f(n) = \log n$

Step 2: Compute ($n^{\log_b a}$)

$$n^{\log_2 1} = n^0 = 1$$

So,

$$n^{\log_b a} = 1$$

Step 3: Compare ($f(n)$)

$$f(n) = \log n$$

Observe that

$$\log n = \Omega(1 \cdot \log^1 n)$$

This corresponds to the generalized Master Theorem where

$$f(n) = \Theta(n^{\log_b a} \log^k n)$$

Here,

- ($k=1$)

Therefore, by the generalized Master Theorem,

$$T(n) = \Theta(\log^{k+1} n)$$

$$T(n) = \Theta((\log n)^2)$$

Recursion Tree Verification

Level 0

Cost

$$\log n$$

Level 1

Cost

$$\log(n/2) = \log n - 1$$

Level 2

Cost

$$\log(n/4) = \log n - 2$$

The recursion continues for

$$\log n$$

levels.

Total cost

$$(\log n) + (\log n - 1) + (\log n - 2) + \dots + 1$$

Using the sum of an arithmetic series,

$$= \log n (\log n + 1) / 2$$

Therefore,

$$\Theta((\log n)^2)$$

Analysis of AI-Based Chatbot Processing

In an AI chatbot:

- User queries are recursively divided into smaller tasks.
- Query understanding, intent detection, and response generation are processed efficiently.
- Only one recursive call is made at each level, reducing computational work.

Because the complexity is:

$$\Theta((\log n)^2)$$

the processing time grows very slowly as the number or size of user queries increases.

Impact on Response Time

- Small increase in execution time even for very large inputs.

- Fast response for thousands or millions of queries.
- Suitable for real-time AI assistants and virtual chatbots.
- Efficient resource utilization and better scalability.